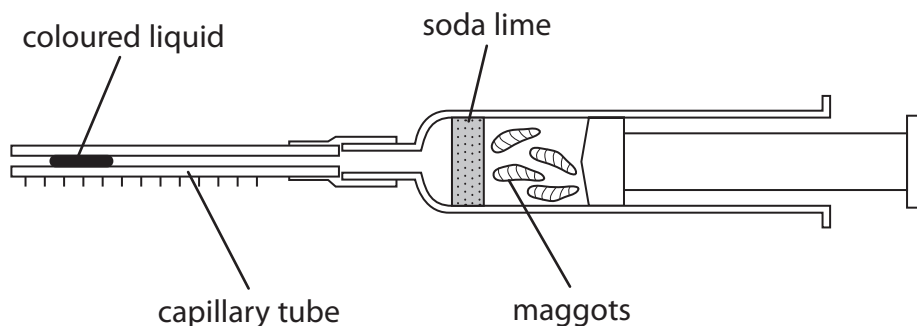


2 The diagram shows some maggots in a respirometer.



This respirometer can be used to investigate the uptake of oxygen during the respiration of maggots.

A student used this respirometer with soda lime to measure the movement of the coloured liquid.

The table shows the results.

minute	1st	2nd	3rd	4th	5th
distance moved (l) by coloured liquid in one minute / mm	4	4	5	4	5

(a) (i) Calculate the **mean** rate of oxygen uptake in $\text{mm}^3 \text{min}^{-1}$.

The internal diameter of the capillary tube was 0.2 mm.

The volume of a cylinder (V) is calculated using the formula

$$\pi r^2 l$$

Give your answer to **two** significant figures.

(3)

..... $\text{mm}^3 \text{min}^{-1}$

(ii) The **mean** rate of carbon dioxide released by the maggots was $0.11 \text{ mm}^3 \text{ min}^{-1}$.

Calculate the respiratory quotient (RQ) value for this investigation.

(2)

RQ

(iii) The soda lime was removed and replaced with glass beads in order to calculate the RQ.

Suggest why this allows the student to calculate the RQ.

(1)

.....

.....

.....

(b) (i) State **two biotic** variables that could affect this investigation.

(2)

First variable

Second variable

(ii) Choose **one** of the variables you have identified in (b)(i).

Describe how this variable could be controlled and the effect it could have on the results if it is not controlled.

(2)

Variable

Describe how this variable is controlled.

.....

.....

.....

Describe the effect it could have on the results if it is not controlled.

.....

.....

.....

(Total for Question 2 = 10 marks)